In [30]:
```python
import pandas as pd
import numpy as np
```

In [31]:
```python
data=pd.read_csv("/content/student_performance_regression_classification.csv")
data
```

Out[31]:

	student_id	pass_final	parental_education	extracurricular	math_score	ma
0	S80000	1.0	Masters	N	57.9	
1	S80001	1.0	Masters	No	73.6	
2	S80002	1.0	High School	es	69.0	
3	S80003	1.0	Bachelors	NO	80.7	
4	S80004	1.0	BACHELORS	Yes	59.2	
...	...	...	...	...	...	
255	S80139	1.0	PHD	No	NaN	
256	S80158	0.0	Phd	YES	76.9	
257	S80238	1.0	BACHELORS	YES	64.9	
258	S80026	0.0	achelors	NO	69.6	
259	S80085	1.0	Masters	Yes	52.3	

260 rows × 12 columns

This dataset contains student information including ID, final exam status, parental education, extracurricular involvement, various math scores, reading score, attendance, study hours, writing score, and calculated total scores. It can be used to analyze factors influencing student performance.

In [32]:
```python
data.isnull().sum()
```

Out[32]:

	0
student_id	19
pass_final	20
parental_education	16
extracurricular	7
math_score	14
math_score.1	14
reading_score	19
attendance_pct	16
study_hours_per_week	11
math_score.2	14
math_score.3	14
writing_score	14

dtype: int64

In [33]:

```
data['math_score']=data['math_score'].fillna(data['math_score'].mean())
data['math_score.1']=data['math_score.1'].fillna(data['math_score.1'].mean())
data['math_score.2']=data['math_score.2'].fillna(data['math_score.2'].mean())
data['math_score.3']=data['math_score.3'].fillna(data['math_score.3'].mean())
data['study_hours_per_week']=data['study_hours_per_week'].fillna(data['study_h
data['writing_score']=data['writing_score'].fillna(data['writing_score'].mean(
data['reading_score']=data['reading_score'].ffill()
data['attendance_pct']=data['attendance_pct'].bfill()
data['extracurricular']=data['extracurricular'].fillna(data['extracurricular']
data['parental_education']=data['parental_education'].fillna(data['parental_ec
data.isnull().sum()
```

Out[33]:

	0
student_id	19
pass_final	20
parental_education	0
extracurricular	0
math_score	0
math_score.1	0
reading_score	0
attendance_pct	0
study_hours_per_week	0
math_score.2	0
math_score.3	0
writing_score	0

dtype: int64

In [34]:

```
data=data.drop(data[data['student_id'].isnull()].index)
data['Total'] = round(data['math_score'] + data['reading_score'] + data['writi
data.loc[data['pass_final'].isnull(), 'pass_final'] = data.loc[data['pass_fina
data.isnull().sum()
```

Out[34]:

	0
student_id	0
pass_final	0
parental_education	0
extracurricular	0
math_score	0
math_score.1	0
reading_score	0
attendance_pct	0
study_hours_per_week	0
math_score.2	0
math_score.3	0
writing_score	0
Total	0

dtype: int64

In [35]:

```
data['extracurricular'] = data['extracurricular'].str.strip().str.lower().repl
print(data['extracurricular'].unique())
data['parental_education']=data['parental_education'].str.strip().str.lower().
    'masters':'Masters', 'master':'Masters', 'mastrs':'Masters', 'msters':'Mas
    'high school':'High School', 'high chool':'High School', 'hgh school':'Hig
    'bachelors':'Bachelors', 'bachlors':'Bachelors', 'bachelor':'Bachelors', '
    'phd':'PhD', 'hd':'PhD', 'pd':'PhD', 'ph':'PhD'
})
print(data['parental_education'].unique())
```

```
['No' 'Yes']
['Masters' 'High School' 'Bachelors' 'PhD' 'achelors']
```

In [36]:

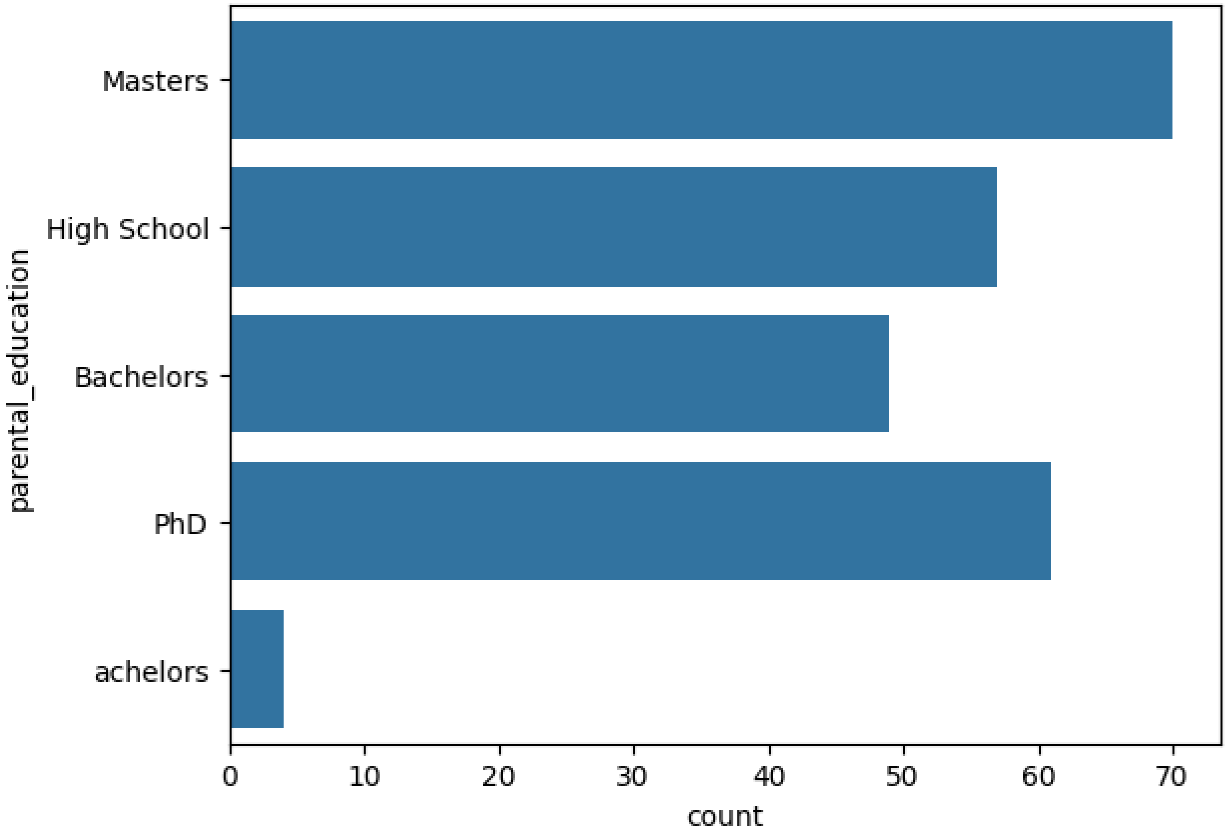
```
data.to_csv('cleaned_dataset.csv', index=False)
```

In [37]:

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.countplot(data['parental_education'])
```

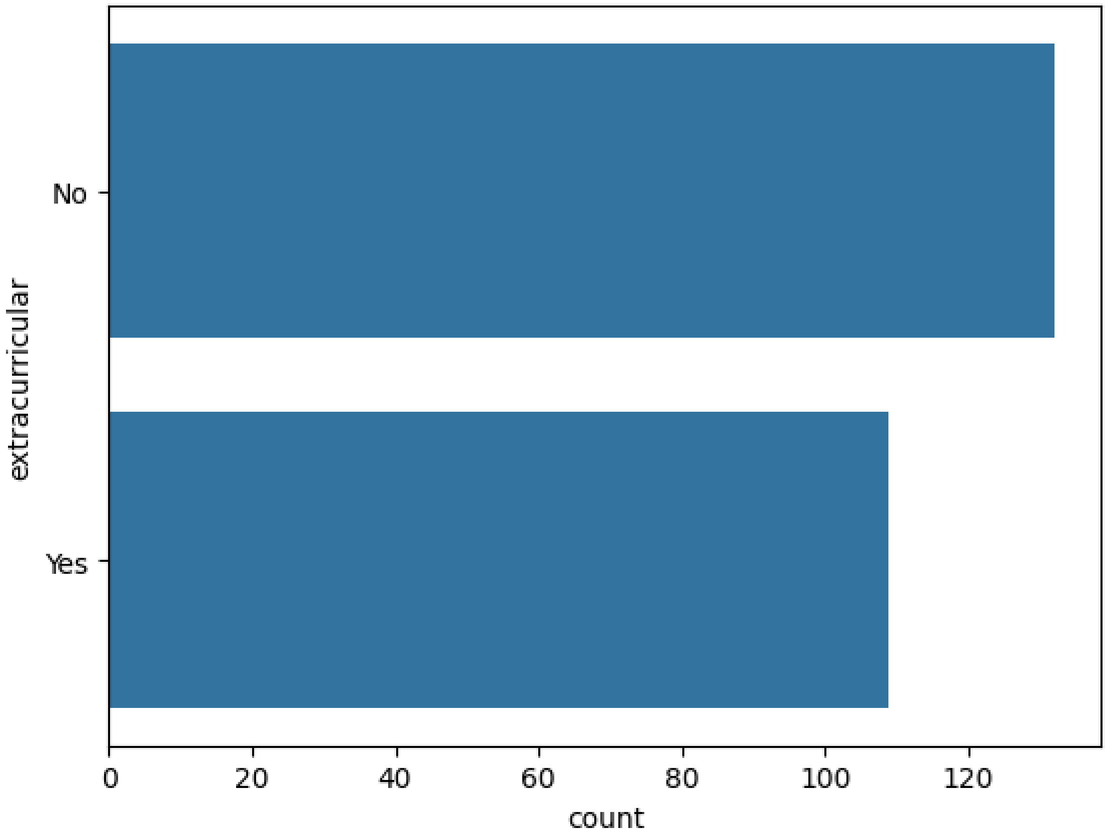
Out[37]:

```
<Axes: xlabel='count', ylabel='parental_education'>
```




```
In [38]: sns.countplot(data['extracurricular'])
```

```
Out[38]: <Axes: xlabel='count', ylabel='extracurricular'>
```



Count Plot:(extra_curricular,parental_education)

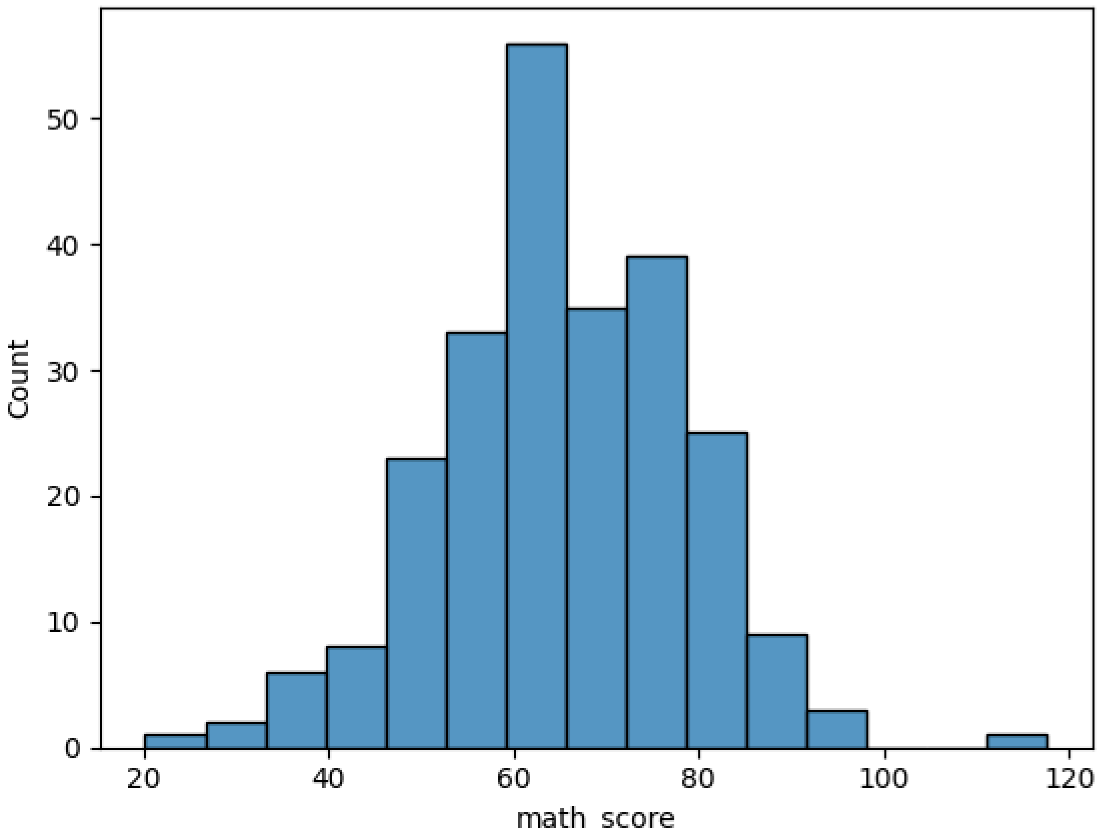
Top education levels: The most common entries are “Masters”, “High School”, and “Bachelors.” That means most parents in this dataset have completed either high school or higher education like college or a master’s degree.

Moderate representation: “PhD” (and its various spellings) appears in the middle range — fewer parents have doctoral degrees compared to Masters or Bachelors.

Rare entries: Categories like “PD”, “Ph”, “Msters”, or “Bachelos” appear only a handful of times. These are likely just typos.

In [39]:
```
sns.histplot(data['math_score'],bins=15)
```

Out[39]:
```
<Axes: xlabel='math_score', ylabel='Count'>
```



Most students scored around 55–70: The tallest bars are in this range, which means most students got math scores around the average level. This suggests that the dataset has a central peak near the mid-60s.

Distribution shape: The histogram looks like a bell curve — wider in the middle and tapering off at both ends. That indicates the math scores follow an approximately normal distribution, meaning:

A few students scored very low (below 40).

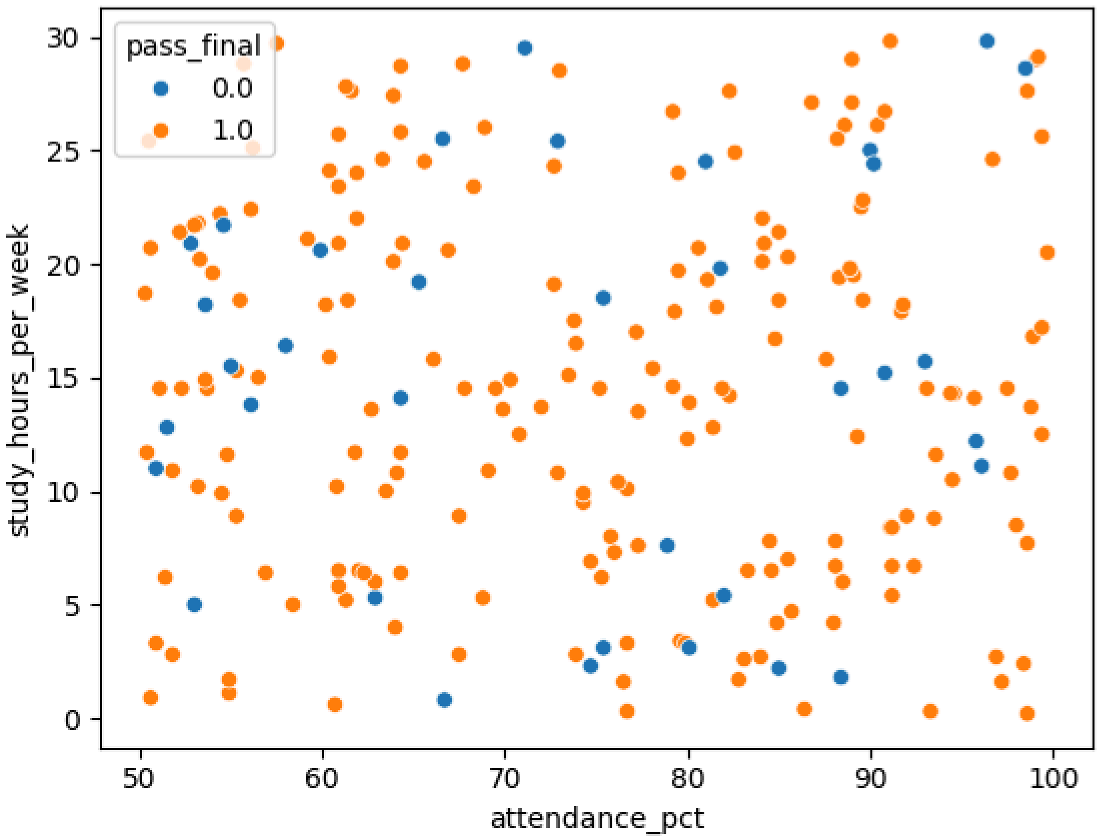
A few scored very high (above 90).

Most are clustered near the mean (60–70).

Outliers: There are very few students who scored above 100 or below 30.

In [40]:
```
sns.scatterplot(x=data['attendance_pct'],y=data['study_hours_per_week'],hue=da
```

Out[40]:
```
<Axes: xlabel='attendance_pct', ylabel='study_hours_per_week'>
```



Clear clustering by color: You can see that most of the orange points (1.0) — the students who passed — are associated with higher scores.

Blue points (0.0) — the students who failed — are more common in the lower score range.

Score threshold effect: There seems to be a rough cutoff point: as the score increases, the likelihood of being marked as pass_final = 1.0 also increases.

This likely reflects the pass mark threshold — for example, students scoring above ~28 or 30 were marked as pass.

Distribution consistency: The points are well spread out across the score range, showing that there's good data variety.

However, since the y-axis doesn't represent a continuous variable, this plot mainly helps to visualize class separation (pass vs fail).

Parental Education: Most parents hold High School, Bachelor's, or Master's degrees — showing a generally educated background.

Student Performance: Most students scored between 55–70, suggesting moderate performance across the group.

Study Hours Impact: Students studying more hours per week tend to pass more consistently, indicating a clear link between study effort and success.